
Low-Resource Emotional TTS via RL

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1 Project Overview

1.1 Profile

Track: 3. Application

Abstract and project goal: We address the problem of emotional capability loss when an expressive TTS model trained on a high-resource language is adapted to a low-resource language. Although the original model can generate emotionally expressive speech, its emotional controllability and prosodic richness often degrade after fine-tuning on limited target-language data. To tackle this issue, we propose an RL-enhanced adaptation framework built upon IndexTTS2 to preserve and realign emotional expressiveness during low-resource language transfer [Zhou et al., 2025]. Our method aims not to learn emotion from scratch, but to retain the emotional prior of the pretrained expressive model while improving emotion intensity, emphasis control, and speech naturalness in the low-resource language. A key component of our approach is a *VA-and-stress control* mechanism that conditions generation on valence–arousal (VA) signals derived from Aspect Sentiment Quad Prediction (ASQP) and injects lexical emphasis markers to improve fine-grained prosodic expressiveness. In our study, we instantiate this setting using Chinese and English as the high-resource source languages and Taiwanese as the low-resource target language.

TL;DR: This project studies how to preserve and recover the emotional expressiveness of an expressive TTS model when adapting it from high-resource languages (Chinese and English) to a low-resource language (Taiwanese), combining RL-based emotional realignment with a VA-and-stress control mechanism to improve emotion intensity and prosodic control.

1.2 Motivation

Expressive speech synthesis is increasingly important for applications such as dubbing, broadcasting, and interactive media. However, when a strong expressive TTS model trained on a high-resource language is adapted to a low-resource language, the model often retains intelligibility but loses emotional expressiveness and fine-grained prosodic control. This makes low-resource adaptation not only a language transfer problem, but also an emotion preservation and alignment problem.

- **Why is the problem interesting?** This problem is interesting because it lies at the intersection of low-resource language adaptation, cross-lingual transfer, and controllable expressive speech synthesis. Rather than learning emotional speech generation from scratch, we focus on preserving and realigning the expressive capability that already exists in the pretrained TTS model when transferring it to a low-resource language.
- **Challenge 1: Emotional Forgetting During Low-Resource Adaptation.** When an emotionally expressive TTS model trained on a high-resource language is fine-tuned on limited low-resource language speech data, it often suffers from catastrophic forgetting, causing the synthesized speech to become flatter and less expressive.

- **Challenge 2: Loss of Fine-Grained Prosodic Control.** After low-resource adaptation, the model may still produce intelligible speech, but fail to preserve natural rhythm, pause structure, emphasis, and emotional intensity that are essential for expressive speech generation.
- **Challenge 3: Difficulty of Cross-Lingual Emotion Alignment.** Emotional cues and prosodic patterns do not transfer perfectly across languages. It is therefore difficult to preserve the expressive prior learned from a high-resource language and align it with the pronunciation, prosody, and linguistic structure of a low-resource language using only limited target-language data.
- **Why the problem remains open or unsolved?** This problem remains open because low-resource languages lack sufficient emotionally annotated speech data for conventional supervised training. As a result, naive fine-tuning often prioritizes pronunciation adaptation over expressive preservation. While seminal foundational models like F5-TTS [Chen et al., 2025] achieve high fidelity, they still lack fine-grained emotional controllability. Conversely, works like EmoSphere++ [Cho et al., 2024] demonstrate strong emotion control but are primarily designed for high-resource scenarios. Extending these capabilities to low-resource languages without catastrophic forgetting remains highly challenging. Several prominent recent works have attempted to enhance expressiveness and language adaptability, including the IndexTTS2 [Zhou et al., 2025] and IndexTTS 2.5 [Li et al., 2026] systems, CosyVoice 3 [Du et al., 2025], and EMORL [Li et al., 2025]. Despite these advances, maintaining deep emotional controllability and prosodic richness during low-resource adaptation without losing the pretrained expressive prior is still an unsolved problem.
- **State-of-the-Art (SOTA) Method:** Currently, IndexTTS2 provides a strong foundation for zero-shot and controllable speech synthesis [Zhou et al., 2025]. In this project, we use it as the expressive base model and investigate how reinforcement learning can help preserve and realign its emotional capabilities during adaptation from a high-resource language to a low-resource language.

2 Problem Formulation

We formulate emotional preservation and prosodic realignment during low-resource language adaptation as a fully observable Markov Decision Process (MDP). Operating within an autoregressive Text-to-Semantic (T2S) environment, the RL agent (a language model policy π_θ) observes a state s encompassing available conditioning information: an input text sequence, a speaker timbre embedding, a duration constraint embedding, and a target global emotion embedding. The action a is the generation of a discrete semantic token sequence, which is subsequently converted into an acoustic waveform. Upon generating a complete sequence, the RL learner receives a multi-objective scalar reward signal $R(s, a)$ computed from the decoded speech. This composite feedback vector, formalized as $R = \lambda_1 R_{\text{emo-cls}} + \lambda_2 R_{\text{intensity}} + \lambda_3 R_{\text{emphasis}} + \lambda_4 R_{\text{ASR}}$, balances four task-specific scalar components: categorical emotion classification accuracy, global emotion intensity (distance to a neutral centroid in Valence-Arousal-Dominance space), local emphasis control (relative pitch and energy patterns), and an ASR-based content reward for semantic intelligibility.

The optimization problem seeks to update the policy parameters θ by maximizing the expected multi-objective reward using Group Relative Policy Optimization (GRPO). The maximization objective, $\mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta} [R(s, a) \nabla_\theta \log \pi_\theta(a|s)]$, is subject to a Kullback-Leibler (KL) divergence penalty constraint, $\beta \text{KL}(\pi_\theta(\cdot|s) \| p_{\text{adapt}}(\cdot|s))$. This constraint anchors the policy to a base model p_{adapt} to prevent catastrophic deviation and preserve the pretrained high-resource expressive prior. Our formulation rests on two key assumptions: (1) an initial parameter-efficient adaptation stage successfully aligns the model to the target low-resource language while keeping most expressive knowledge intact, and (2) the discrete semantic token space produced by the T2S module is sufficiently expressive to retain and recover subtle emotional intensities and local prosody variations during final acoustic reconstruction.

3 Empirical Evaluation

To rigorously evaluate the effectiveness of our proposed RL-enhanced IndexTTS2 framework in preserving and recovering emotional expressiveness during transfer from a high-resource language to a low-resource language, we employ a combination of objective and subjective evaluations.

3.1 Performance Metrics

We plan to adopt several key performance metrics to comprehensively assess speech intelligibility, speaker fidelity, emotional expressiveness, and emotion retention after low-resource adaptation.

- **Objective Metrics:**

- **Word Error Rate (WER):** Evaluates speech intelligibility and articulation accuracy using ASR models like FunASR [An et al., 2025] or Whisper [Radford et al., 2022].
- **Speaker Similarity (SS):** Computed as the cosine similarity between speaker embeddings extracted from a pretrained speaker recognition model to ensure the timbre is maintained after low-resource adaptation and RL optimization.
- **Speech Emotion Recognition (SER):** Measured using the open-source emotion2vec model to objectively verify if the generated speech matches the target emotion category.

- **Subjective Metrics (MOS Framework):**

- **Multidimensional MOS:** We will conduct human evaluations based on a 1-5 scale for Similarity MOS ($\uparrow$), Intelligibility MOS ($\uparrow$), and Naturalness MOS ($\uparrow$).

3.2 Baseline Methods

We compare our method against several baselines to explicitly quantify emotional forgetting and emotional recovery during low-resource adaptation.

- **CosyVoice2** [Du et al., 2024]: A strong text-to-speech baseline fine-tuned on the target low-resource language for cross-model comparison.
- **CosyVoice3** [Du et al., 2025]: An improved model built upon CosyVoice2, which is also fine-tuned on the target data to evaluate cross-lingual generalization and speech generation quality.
- **IndexTTS2 + Naive Low-Resource Fine-Tuning:** A baseline that fine-tunes IndexTTS2 on limited target-language data without any explicit preservation mechanism, used to measure the severity of emotional forgetting.
- **IndexTTS2 + FT + GRPO:** A target-language adaptation baseline using a lightweight adaptation strategy, used to test whether preserving more pretrained parameters helps retain emotional expressiveness.
- **IndexTTS2 + FT + GRPO + emphasis + va (Ours):** Our full method, which applies RL-based emotional and prosodic alignment after low-resource adaptation.

3.3 Benchmark Datasets

To train the low-resource adaptation model and evaluate catastrophic forgetting of emotional expressiveness, we employ the following datasets:

- **Taiwanese Across Taiwan Corpus (TAT)** [Liao, 2022]: A NAER-hosted large-scale native Taiwanese speech corpus. We utilize the TAT-MOE subset, containing approximately 200 hours of recordings, as the training set for low-resource fine-tuning. For validation and testing, we use the evaluation and test sets from TAT Volume 1 (TAT-Vol1), each comprising approximately 5 hours of data.

4 Methodology

Our methodology follows a two-stage framework built upon a pre-trained expressive IndexTTS2 model. Starting from a model trained on a high-resource language, we first adapt it to the low-resource target language, and then apply reinforcement learning to preserve and recover emotional expressiveness, prosodic control, and content accuracy during transfer.

4.1 Stage 1: Low-Resource Language Adaptation

In the first stage, we adapt the expressive base model to the low-resource target language. The goal of this stage is to acquire target-language pronunciation and linguistic patterns while preserving as

much of the pretrained expressive prior as possible. To reduce severe catastrophic forgetting, we adopt a parameter-efficient adaptation strategy, such as lightweight adapter modules or LoRA, so that only a limited subset of model parameters is updated during target-language transfer.

4.2 ASQP Annotation Pipeline

The Valence–Arousal (VA) conditioning and local emphasis annotations required for the *VA-and-stress control* are derived from an Aspect Sentiment Quad Prediction (ASQP) model, which operates on Traditional Chinese text. The RL training data is sourced directly from the SemEval-2026 Task 3 Chinese restaurant dataset [DimABSA, 2026], which provides Traditional Chinese sentences with aspect–sentiment quadruple annotations and thus requires no translation.

For the evaluation corpus (TAT-Vol1 Taiwanese test utterances), however, no Chinese-language annotations are available. Since available ASQP tools do not support the romanized Taiwanese representation, we adopt a back-translation pipeline for this portion only: each Taiwanese test sentence is first translated into Traditional Chinese, processed by the ASQP model to extract aspect–sentiment quadruples, and then translated back to Taiwanese. The reverse translation step employs the heterogeneous model ensemble with adaptive weighted voting proposed in our previous work [Hsieh et al., 2026], which aggregates multiple translation candidates to improve lexical fidelity under the low-resource constraint. The resulting sentences, together with their ASQP-derived annotations, form the *Back-Translated Dataset* used to evaluate IndexTTS2-FT-va-emphasis, IndexTTS2-FT+RL, and IndexTTS2-proposed.

4.3 Phonemic Text Front-End

Because the low-resource target language is written with Han characters but pronounced according to its own phonology, we drive the model with a romanized phonemic representation rather than raw Han text. Each input sentence is converted to Taiwanese romanization (Tâi-lô) with numeric tone marks (e.g., gua2 huann1-hi2), and locally emphasized spans are wrapped in quotation markers that the model learns to render as prosodic stress, providing the textual signal that the Local Emphasis Control Reward acts on. Crucially, the backbone’s built-in text normalizer—designed for Mandarin/English number and symbol expansion—is bypassed, since the input is already phonemic and normalization would corrupt both the romanization (including the tone digits) and the emphasis markers. The identical front-end is applied during RL training and inference, so the representation the policy is optimized on exactly matches the one used at generation time.

4.4 Stage 2: RL-Based Emotional Realignment

In the second stage, we introduce an RL-based alignment procedure [Li et al., 2025] on top of the adapted model. The core idea is to treat the autoregressive generation of semantic speech representations as a sequential decision-making process, while computing task-specific rewards from the decoded speech waveform. This stage explicitly optimizes emotion preservation, emotional intensity, local emphasis, and content intelligibility in the low-resource language. In practice, the RL updates are mainly applied to the autoregressive T2S transformer, and can optionally include the emotion- or style-related conditioning modules when stronger expressive realignment is needed.

4.5 Reinforcement Learning Formulation

The generation of semantic speech tokens is formulated as a Markov Decision Process (MDP), while the reward is computed from the decoded speech waveform:

- **State** (S_t): The hidden states of the IndexTTS2 autoregressive decoder, incorporating the input text, speaker conditions, and previously generated speech tokens up to step t .
- **Action** (A_t): The predicted semantic speech token at step t , sampled from the policy distribution $\pi_\theta(A_t|S_t)$.
- **Reward** (R_t): A composite reward signal provided upon the completion of a sequence or segment. The reward is designed to preserve emotional expressiveness while maintaining intelligibility in the low-resource target language, and comprises four components:

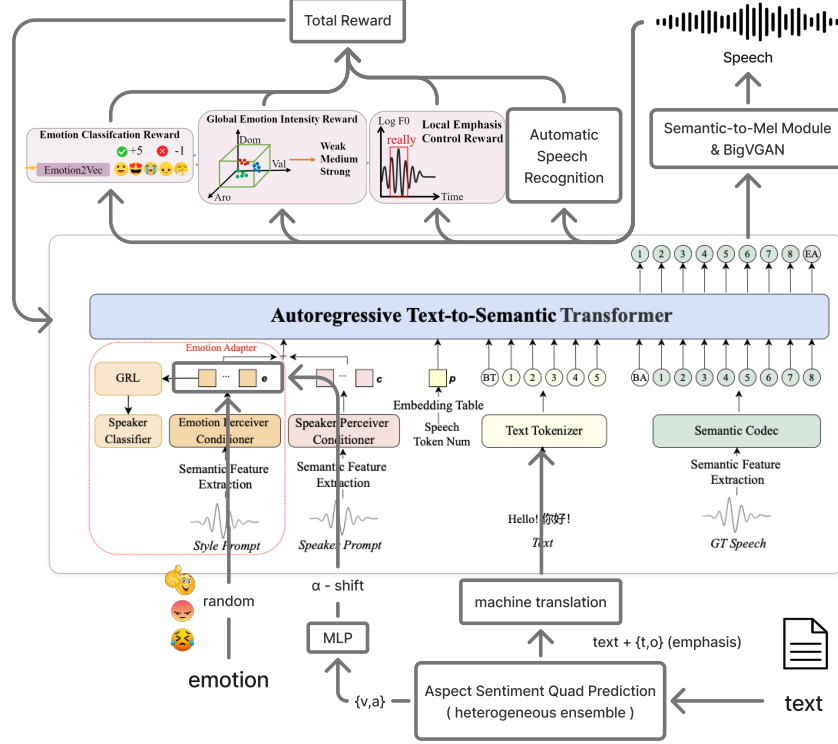


Figure 1: High-level architecture of the proposed RL-enhanced IndexTTS2 framework. After low-resource adaptation, the generator outputs semantic speech tokens, which are decoded into speech waveforms and evaluated by emotion, intensity, prosody, and content reward models. These rewards are used to update the policy via Group Relative Policy Optimization (GRPO), mitigating emotional forgetting during language transfer.

1. *Emotion Classification Reward*: Evaluated using a pre-trained emotion recognition model (e.g., emotion2vec) to measure whether the generated speech matches the target emotion category.
2. *Global Emotion Intensity Reward*: Computed from the distance between the generated speech and a neutral centroid in the Valence-Arousal-Dominance (VAD) space, in order to encourage the target global emotional intensity.
3. *Local Emphasis Control Reward*: A reward based on the relative pitch and energy patterns of emphasized words, which encourages correct placement of local stress and expressive prosody.
4. *ASR-Based Content Reward*: Computed using an ASR model on the generated speech to compare the transcription against the input text, encouraging content preservation and intelligibility during RL optimization.

In summary, the sequence-level reward can be written as

$$R = \lambda_1 R_{\text{emo-cls}} + \lambda_2 R_{\text{intensity}} + \lambda_3 R_{\text{emphasis}} + \lambda_4 R_{\text{ASR}},$$

where the coefficients λ_i control the trade-off between emotional expressiveness, local prosody, and content accuracy.

4.6 Alleviating Catastrophic Forgetting

To address the degradation of emotional capabilities during low-resource fine-tuning, we initialize our policy π_θ with the weights of the target-language adapted IndexTTS2 model rather than training from scratch. We then employ Group Relative Policy Optimization (GRPO) mixed with a Kullback-Leibler (KL) divergence penalty against the adapted reference model. This ensures that the model explores localized emotional and prosodic improvements while preserving the linguistic adaptation and expressive prior inherited from the original high-resource model.

5 Midterm Experimental Results

In this section, we present a toy-example diagnostic experiment designed to illustrate the limitations of the baseline systems, CosyVoice2 and CosyVoice3 (CV2/CV3), on Taiwanese emotional TTS. Rather than serving as a full final evaluation, this experiment uses a controlled synthesis setup to expose the trade-offs between intelligibility, emotion controllability, and speaker fidelity. We compare these baselines with IndexTTS2, our current method in this midterm study, to show that the proposed direction can more effectively handle the observed baseline weaknesses, especially in preserving intelligibility while enabling stronger emotional variation.

5.1 Preliminary Baseline Evaluation

As an initial pilot study, we conducted a subjective evaluation to establish a preliminary baseline for Taiwanese TTS. The evaluation used the Mean Opinion Score (MOS) on a scale from 1 to 5 (where 5 represents the best performance). The evaluation panel consisted of 12 members from the SARC laboratory. The test corpus involved 20 sentences for evaluating speech intelligibility and naturalness, and an additional 10 sentences specifically for evaluating voice similarity. In total, the subjective evaluation yielded 600 rating samples ((20 * 2 + 10) test items × 12 evaluators).

Table 1 shows the preliminary subjective evaluation results of the baseline method, CosyVoice2 [Du et al., 2024], on Taiwanese TTS. The model achieves a similarity MOS of 4.16, an intelligibility MOS of 4.41, and a naturalness MOS of 4.37, resulting in an overall mean MOS of 4.31. These results suggest that a strong off-the-shelf baseline can already generate high-quality Taiwanese speech. This observation motivates our project to move beyond basic intelligibility and naturalness, and to focus on whether emotional expressiveness can be preserved and recovered during low-resource adaptation.

Table 1: Preliminary Subjective Evaluation (MOS) on Taiwanese TTS.

Model	Similarity MOS ↑	Intelligibility MOS ↑	Naturalness MOS ↑	Mean ↑
CosyVoice2	4.16	4.41	4.37	4.31

5.2 Evaluation Synthesis Protocol

To compare emotional expressiveness across models on Taiwanese, we synthesized a large-scale evaluation corpus from the TAT-Vol1 texts (2,752 utterances). All systems were conditioned on the same fixed reference speaker audio (a single Taiwanese speaker) to control for speaker identity; only the emotion label was varied across the seven target categories: *angry*, *happy*, *fearful*, *disgusted*, *sad*, *surprised*, and *neutral*. Emotion labels were assigned in a balanced, shuffled order across utterances (random seed fixed at 42 for reproducibility). For the IndexTTS2 systems, prompts are synthesized from Taiwanese romanized text with the built-in text normalizer disabled, matching the phonemic front-end used during training (Section 4). Table 2 summarizes the inference setup and generation statistics for each system.

Table 2: Evaluation synthesis statistics for the three TTS models on 2,752 TAT-Vol1 utterances.

Model	Emotion Control	Audio Duration	Failures
CosyVoice2	Text prompt	07:53:41	39
CosyVoice3	Text prompt	09:48:08	5
IndexTTS2	8-dim vector	09:21:57	0
Total		27:04:06	44

5.3 Speech Emotion Recognition (SER)

To evaluate whether the intended emotion is recognizable in the synthesized speech, we apply the emotion2vec speech emotion recognition (SER) model to the generated utterances and report the per-emotion recognition accuracy together with the corpus-wide mean. For IndexTTS2 we evaluate a 2 × 2 ablation that crosses the *training stage* with the *generation-time control*. The two checkpoints

are the low-resource fine-tuned model (before RL) and the same model after our RL realignment; on each checkpoint we render emotion either with a plain fixed-magnitude 8-dimensional emotion vector, or with our *VA-and-stress control*, which (i) scales that emotion vector by a valence–arousal (VA) conditioned adjustment learned during training and (ii) injects lexical stress by wrapping the sentiment-bearing spans in emphasis markers. This yields four systems: **IndexTTS2-FT** (fine-tuned, no control), **IndexTTS2-FT+va-emphasis** (fine-tuned + control), **IndexTTS2-FT+RL** (RL, no control), and **IndexTTS2-proposed** (fine-tuned + RL + control, our full system). The two fine-tuned rows and the two RL rows each share an *identical* checkpoint and differ only in generation-time rendering, so contrasting them isolates exactly what the VA-and-stress control contributes; contrasting the fine-tuned rows against the RL rows isolates the effect of the RL stage. Results are summarized in Table 3.

Table 3: Speech Emotion Recognition (SER) accuracy broken down by target emotion (higher is better, $\uparrow$).

Model	Mean $\uparrow$	Neutral	Angry	Happy	Sad	Surprise	Fear	Disgust
CosyVoice2	0.14	0.96	0.00	0.02	0.02	0.00	0.00	0.01
CosyVoice3	0.13	0.67	0.00	0.05	0.14	0.02	0.00	0.03
IndexTTS2-FT	0.27	0.94	0.22	0.49	0.01	0.20	0.01	0.00
IndexTTS2-FT+va-emphasis	0.27	0.95	0.19	0.52	0.01	0.18	0.01	0.01
IndexTTS2-FT+RL	0.28	0.93	0.26	0.41	0.00	0.34	0.00	0.01
IndexTTS2-proposed	0.30	0.93	0.20	0.58	0.00	0.35	0.01	0.01

As shown in Table 3, all four IndexTTS2 systems clearly outperform the CosyVoice baselines (0.14 and 0.13) on mean recognition accuracy. The central observation is that the benefit of the VA-and-stress control becomes substantially more apparent after RL realignment. On the fine-tuned checkpoint, adding the control leaves the mean unchanged (0.27 $\rightarrow$ 0.27): a gain on *happy* (0.49 $\rightarrow$ 0.52) is offset by small drops on *anger* (0.22 $\rightarrow$ 0.19) and *surprise* (0.20 $\rightarrow$ 0.18). On the RL checkpoint the *same* control raises the mean from 0.28 to 0.30, driven by an increase on *happy* (0.41 $\rightarrow$ 0.58) with *surprise* held at $\approx$ 0.34; here too *anger* is the only clear regression (0.26 $\rightarrow$ 0.20), suggesting the emphasis and VA scaling occasionally trade anger-specific cues for the larger expressiveness gain on high-valence emotions. RL realignment on its own moves the mean only modestly (0.27 $\rightarrow$ 0.28) but reshapes the profile—boosting *surprise* (0.20 $\rightarrow$ 0.34) and *anger* (0.22 $\rightarrow$ 0.26)—suggesting that RL realignment increases the effectiveness of the VA-and-stress control. *Sad*, *fear* and *disgust* remain near zero for every system, suggesting that these emotions remain difficult to distinguish under the current SER-based evaluation.

5.4 Word Error Rate (WER)

We transcribe each synthesized utterance using a locally hosted `faster-whisper-taigi-pinyin-large-v7` model, a Whisper-large checkpoint fine-tuned for Taiwanese pinyin transcription. Both the reference (TAT-Vol1) text and the ASR hypothesis are normalized before scoring, and we consider two normalization settings that differ in how tones are handled. For the CosyVoice2 and CosyVoice3 baselines, WER is computed in a *tone-sensitive* setting: hyphens and punctuation (`- . , ! ?`) are stripped, whitespace is collapsed, and both strings are lowercased, but the per-syllable tone numbers are *retained*, so a syllable counts as correct only when both its romanization and its tone match. For the Ground Truth reference and our IndexTTS2 system, we additionally strip the tone numbers (together with the remaining special symbols) before scoring, yielding a *tone-insensitive* WER that forgives tone-only errors. Because tone-sensitive scoring is strictly harder, the larger error rates reported for CV2/CV3 are partly attributable to this stricter normalization; the absolute WER values should therefore be read as a diagnostic proxy rather than as directly comparable intelligibility scores. Word-level edit-distance statistics (hits, substitutions, deletions, insertions) are computed via the `jwer` library, and we report the average per-utterance WER; for analysis and debugging, the output CSV keeps both the original and the normalized reference/hypothesis strings.

We deliberately break the WER down by target emotion in order to examine whether injecting different emotions degrades the textual intelligibility of the synthesized speech: a model that preserves clarity should yield a roughly flat WER profile across emotions, whereas large emotion-dependent swings would indicate that strong emotional rendering harms pronunciation. Failed-generation cases (*Missing*) are excluded from WER computation and are analyzed separately. The Ground Truth

row corresponds to the original reference sentences, which are not produced by any TTS model and therefore carry no injected emotion; accordingly we report only its overall Mean—which captures the residual error of the ASR system itself and serves as a reference point for the baseline ASR error—and leave the per-emotion columns blank. Results are summarized in Table 4.

Table 4: Word Error Rate (WER) across the full synthesized corpus and broken down by target emotion (lower is better, ↓).

Model	Mean ↓	Neu.	Ang.	Hap.	Sad	Sur.	Fea.	Dis.
<i>Original Dataset</i>								
Ground Truth	0.3989	-	-	-	-	-	-	-
CosyVoice2	1.6663	1.6392	1.6884	1.6565	1.6570	1.6879	1.6483	1.6865
CosyVoice3	2.1102	2.1140	2.1374	2.0907	2.0798	2.1003	2.1413	2.1077
IndexTTS2-FT	0.7050	0.6610	0.7285	0.7383	0.6103	0.6435	0.8222	0.7313
<i>Back-Translated Dataset</i>								
IndexTTS2-FT+va-emphasis	0.7408	0.8618	0.6672	0.5962	0.6899	0.6585	0.8153	0.8972
IndexTTS2-FT+RL	0.8162	0.9065	0.8289	0.9268	0.8237	0.6253	0.9585	0.6434
IndexTTS2-proposed	0.8404	0.8453	0.7707	0.8940	0.8031	0.7852	0.9718	0.8132

All four IndexTTS2 systems are scored under the same faster-whisper-taigi-pinyin ASR model and tone-insensitive normalization, so their WER values are directly comparable; the CosyVoice rows use a different ASR model and stricter tone-sensitive normalization and therefore remain a within-pipeline diagnostic proxy rather than directly comparable numbers. The table is organized by training data into two groups. The *Original Dataset* group contains the Ground Truth reference, both CosyVoice baselines, and IndexTTS2-FT; within this group, IndexTTS2-FT achieves the lowest WER across all emotion categories (Mean: 0.7050), substantially outperforming both CosyVoice baselines. The *Back-Translated Dataset* group contains the three remaining IndexTTS2 variants; within this group, IndexTTS2-FT+va-emphasis records the lowest mean WER (0.7408) and leads on most individual emotions, IndexTTS2-FT+RL is best on Sur (0.6253) and Dis (0.6434), and IndexTTS2-proposed achieves the lowest WER on Neu (0.8453). Across both groups, WER increases monotonically along the pipeline: 0.7050 (FT) → 0.7408 (FT+va-emphasis) → 0.8162 (FT+RL) → 0.8404 (proposed); however, since the back-translated variants also differ from IndexTTS2-FT in training data, this trend reflects a combination of component additions and dataset change, and no single-factor attribution can be made from these results alone. The VA-and-stress control contributes a relatively small and consistent cost on *both* checkpoints (FT: +0.036; RL: +0.024), while the RL stage itself accounts for the largest single increment (0.7050 → 0.8162, +0.111, approximately +15.7% relative). Overall, the strongest emotional rendering (proposed) raises WER by 0.135 absolute over the plain fine-tuned model—a non-negligible but bounded degradation—suggesting that neither RL realignment nor the VA-and-stress control substantially impairs pronunciation, and that the emotional gains in Table 3 are achieved with only a modest intelligibility cost, indicating a favorable trade-off between emotional expressiveness and intelligibility.

5.5 Speaker Similarity (SS)

Speaker embeddings are extracted using SpeechBrain’s ECAPA-TDNN encoder pre-trained on VoxCeleb (speechbrain/spkrec-ecapa-voxceleb). Each audio file is mono-mixed and resampled to 16 kHz before being encoded. Speaker Similarity (SS) is then defined as the cosine similarity between the embedding of the synthesized audio and that of a fixed reference recording (a single Taiwanese speaker), so that SS measures how well each model preserves the target timbre across all synthesized emotional variants. Because emotional rendering perturbs the acoustic features used by the speaker encoder, we report both the corpus-wide mean and the emotion-wise breakdown in Table 5.

Table 5 exposes the timbre cost of stronger emotional control, and it mirrors the SER pattern in a revealing way. All IndexTTS2 systems preserve the target timbre markedly less well than the CosyVoice baselines (around 0.57–0.58). Crucially, the VA-and-stress control behaves *oppositely* on the two checkpoints: on the fine-tuned model it leaves speaker similarity essentially untouched—in fact nudging it slightly *up*, 0.4561 → 0.4633, uniformly across categories—whereas on the RL

Table 5: Speaker Similarity (SS) across the full synthesized corpus and broken down by target emotion (higher is better, $\uparrow$).

Model	Eval.	Mean $\uparrow$	Neu.	Ang.	Hap.	Sad	Sur.	Fea.	Dis.
CosyVoice2	2713	0.5812	0.5877	0.5864	0.5761	0.5903	0.5773	0.5711	0.5793
CosyVoice3	2747	0.5725	0.5699	0.5749	0.5747	0.5718	0.5698	0.5715	0.5751
IndexTTS2-FT	2752	0.4561	0.5675	0.2940	0.4809	0.4557	0.4118	0.4899	0.4934
IndexTTS2-FT+va-emphasis	2752	0.4633	0.5827	0.3000	0.4844	0.4630	0.4182	0.4977	0.4976
IndexTTS2-FT+RL	2752	0.4333	0.5604	0.2737	0.4474	0.4251	0.3835	0.4725	0.4708
IndexTTS2-proposed	2752	0.4155	0.5744	0.2486	0.4346	0.4002	0.3636	0.4383	0.4494

checkpoint it *lowers* similarity, $0.4333 \rightarrow 0.4155$, again consistently across every emotion (e.g. anger $0.2737 \rightarrow 0.2486$, fear $0.4725 \rightarrow 0.4383$, sadness $0.4251 \rightarrow 0.4002$), sparing only the timbre-stable *neutral* case. This pattern is broadly consistent with the SER results: the control markup has a limited observable effect on the fine-tuned model—changing neither emotion nor timbre noticeably—and shows stronger effects after RL realignment, at which point it simultaneously sharpens emotion (Table 3) and coincides with lower speaker similarity scores, with the largest degradation on high-arousal *anger*. The RL stage alone also lowers similarity ($0.4561 \rightarrow 0.4333$), so along the no-control $\rightarrow$ RL $\rightarrow$ proposed path overall similarity falls monotonically ($0.4561 \rightarrow 0.4333 \rightarrow 0.4155$).

Conclusion. Our IndexTTS2 results form a clean 2×2 ablation over the training stage (fine-tuned vs. RL-realigned) and the generation-time control (none vs. VA-and-stress). The dominant finding is that the effect of the VA-and-stress control becomes substantially more apparent after RL realignment. On the fine-tuned checkpoint the control has a limited observable effect: it leaves SER unchanged ($0.27 \rightarrow 0.27$, Table 3) and even nudges speaker similarity slightly up ($0.4561 \rightarrow 0.4633$, Table 5). After RL realignment the *same* control shows substantially stronger effects, raising SER to 0.30—driven mainly by *happy* ($0.41 \rightarrow 0.58$)—but now accompanied by a speaker-similarity reduction ($0.4333 \rightarrow 0.4155$). Within the RL-aligned systems, improvements in emotion recognition are accompanied by reductions in speaker similarity. Relative to the CosyVoice baselines, this places IndexTTS2 at the opposite end of the same trade-off: it recognizes emotion far better (0.30 vs. 0.13–0.14) but preserves timbre less well ($SS \approx 0.42$ – 0.46 vs. ≈ 0.57).

For intelligibility, all four IndexTTS2 systems are now scored under the same ASR model and tone-insensitive normalization pipeline and are therefore directly comparable; WER rises monotonically along the pipeline ($0.7050 \rightarrow 0.7408 \rightarrow 0.8162 \rightarrow 0.8404$), with the RL stage contributing the largest single increment and the VA-and-stress control adding a small, consistent cost on either checkpoint. The CosyVoice rows use a different ASR model and a stricter tone-sensitive normalization, so that part of the WER column remains a within-pipeline diagnostic proxy rather than a cross-model ranking. Overall, our pipeline progressively strengthens emotional controllability with a bounded increase in WER; a key remaining challenge—and the focus of the next stage—is improving speaker similarity while retaining the emotion-recognition gains observed after RL realignment and VA-and-stress control.

6 Expected Contributions of Each Team Member

- **Hao-Chun Hsieh:** Leads the overall project direction, problem formulation, and methodology design. He is also responsible for developing the base codebase, establishing the theoretical foundations, and drafting the project proposal.
- **Pao-Hua Hsu:** Generates the required audio files and refines the project proposal. Also supporting system architecture diagrams and designing the final poster.
- **Nien-Tse Wu:** Extends the base codebase to ensure a lightweight and highly efficient implementation. He also refines the project proposal alongside spearheading empirical evaluation, dataset curation, and hyperparameter tuning.

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